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Asymmetric Cryptography

Data Privacy and Security

Data Science and Management (DASMA)

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Issues with Symmetric Cryptography

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   - They could exchange it physically, but it is not practical for large-scale deployments.

2. **Key Management**, i.e., when we have n users, we need to manage $n(n-1)/2$ keys:
   - There are billions of users accessing the internt, it cannot scale!
   - For instance, Amazon has over 100M users and would need to manage $100M \times 100M/2 = 50$ billion keys!
   - Even small organizations have thousands of users and may struggle to manage a their keys.

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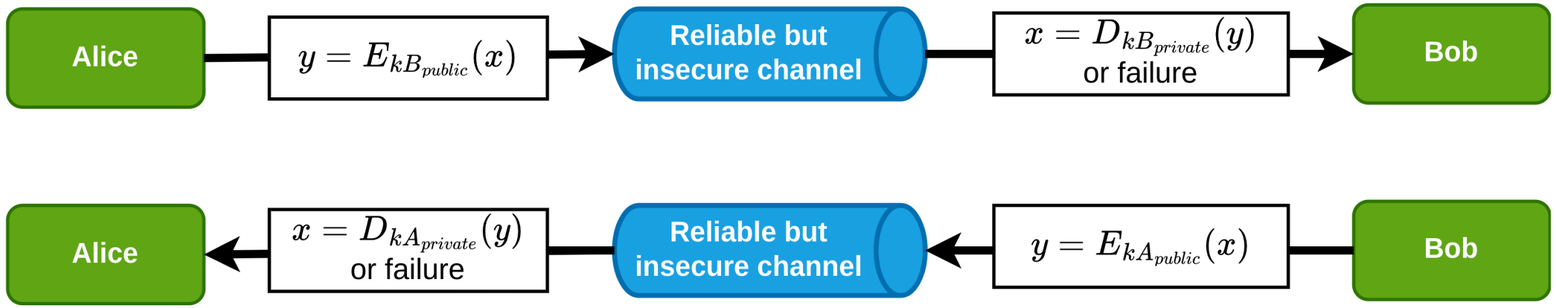
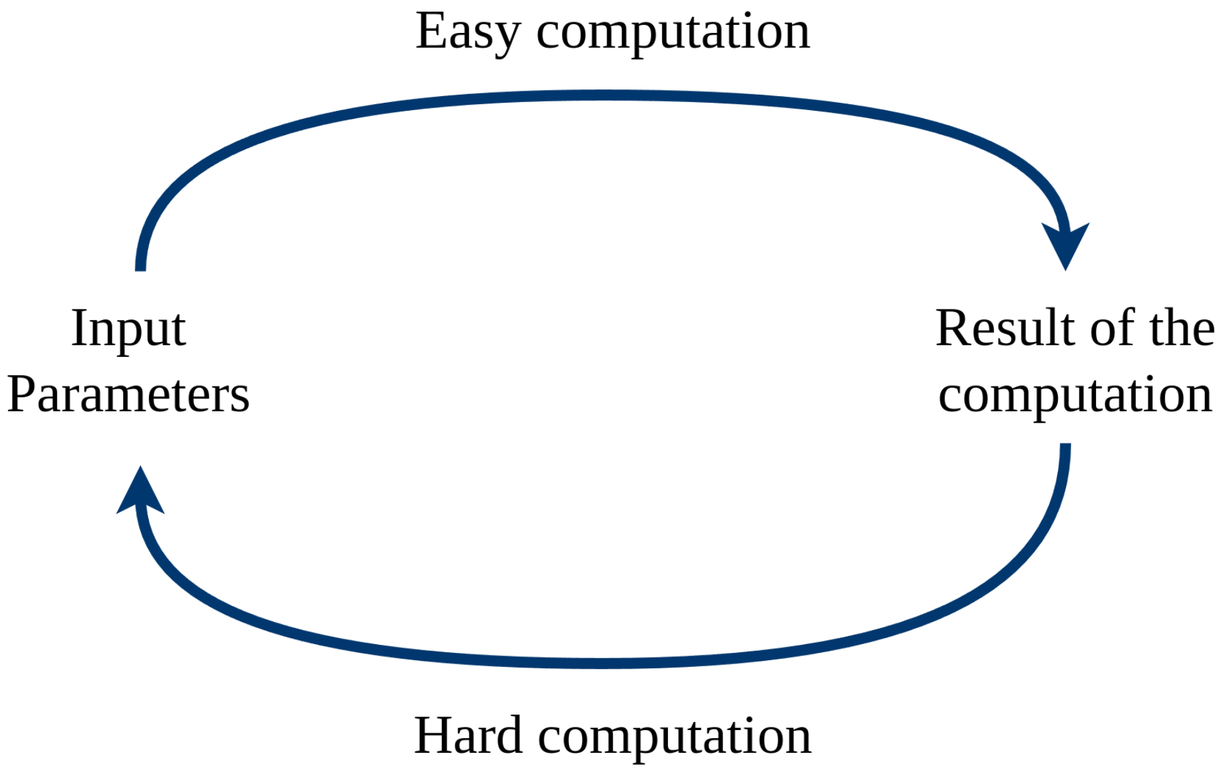
Hence:



where $kA_{private}$ is the private key of Alice and kA_{public} is the public key of Alice, similarly $kB_{private}$ is the private key of Bob and kB_{public} is the public key of Bob.

How to achieve this *asymmetry*?

Most methods exploit a mathematical problem showing this behavior:



Easy to compute the result starting from the input parameters
but hard to compute the input parameters starting from the result

Diffie and Hellman Algorithm for Key Exchange

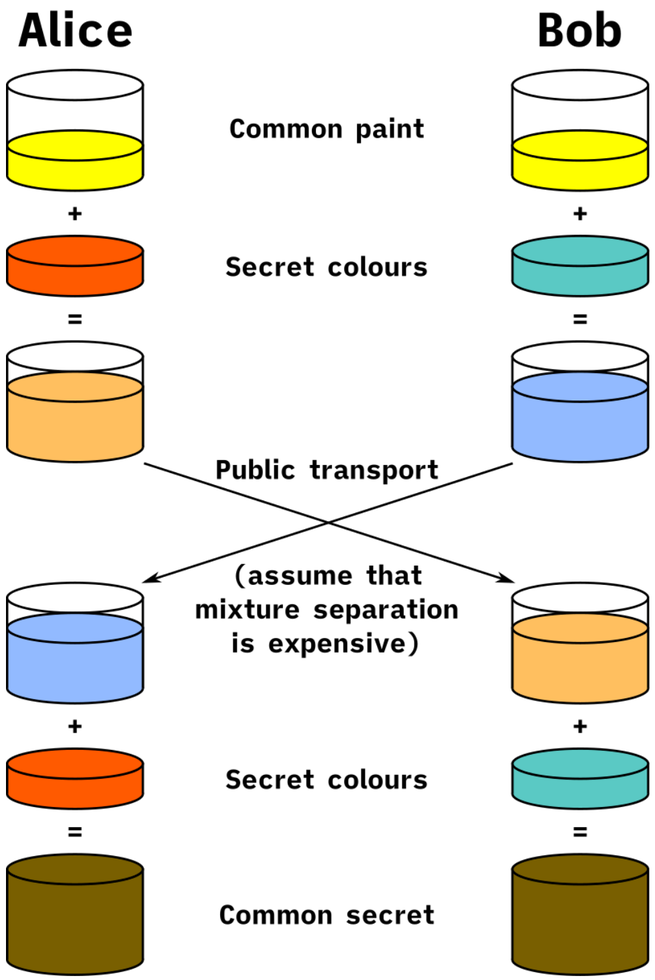
Key Exchange

Alice and Bob wants to encrypt their messages using symmetric cryptography:

- They need to exchange the symmetric key
- They don't know each other and cannot meet in person
- They cannot exchange the key over an insecure channel
- They want to agree on the key without revealing it to an eavesdropper
- They can exchange messages over an insecure channel but not the key

Intuition of Diffie-Hellman Key Exchange Algorithm

Visually*:

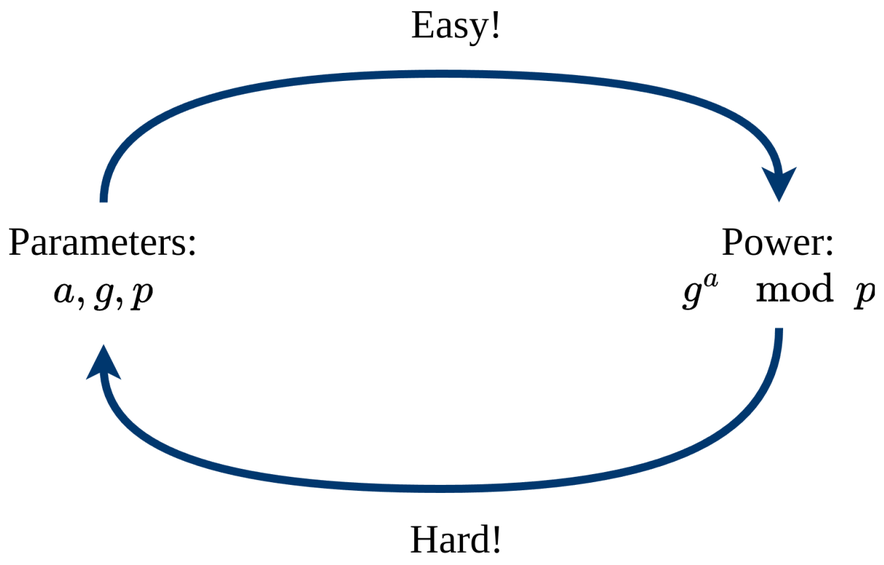


* In this example, we are assuming that mixture separation is hard to perform. It is not the case in real life.

Instead of colors we use power of numbers

Diffie-Hellman key exchange protocol exploits the Discrete Logarithm Problem (DLP):

- Given a, p, and g, it is easy to compute $g^a \mod p$
- Given the result of $g^a \mod p$, it is hard to compute a even when p and g are known.



REMARK: $x \mod p$ operation means taking the remainder of the division of x by p.

But computing the logarithm is easy!

Given a **real** number, any calculator can compute:

$$\log_g(g^a) = a \qquad \Longrightarrow \qquad \log_7(7^8) = \log_7(5764801) = 8$$

Indeed, if g is known, the larger the value of g^a, the larger the value of a will be:

- $7^0 = 1$
- $7^1 = 7$
- $7^2 = 49$
- $7^3 = 343$
- $7^4 = 2401$
- $7^5 = 16807$

The logarithm is a *monotonic* function and we can find its approximate value by hand with *some* attempts*.

* We are oversimplifying the problem to get the intuition. However, there exists numerical methods to quickly get the solution.

But moving to the discrete world changes everything

However, when we move to the **discrete** logarithm, we do not know how to do it efficiently:

- Even if g and p are known, as $g^a \mod p$ increases, a does **not** necessarily increase.

- Assuming $g = 7, p = 97$, then:
  - $7^0 \mod 97 = 1 \mod 97 = 1$
  - $7^1 \mod 97 = 7 \mod 97 = 7$
  - $7^2 \mod 97 = 49 \mod 97 = 49$
  - $7^3 \mod 97 = 343 \mod 97 = 52$
  - $7^4 \mod 97 = 2401 \mod 97 = 73$
  - $7^5 \mod 97 = 16807 \mod 97 = 26$
  - $7^6 \mod 97 = 117649 \mod 97 = 85$

Diffie-Hellman Key Exchange Algorithm (1976)

Public parameters:

- Prime number p
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$g^a \mod p$		$g^b \mod p$
	$\xrightarrow{g^a \mod p}$	
	$\xleftarrow{g^b \mod p}$	
$(g^b)^a \mod p = g^{ab} \mod b$		$(g^a)^b \mod p = g^{ab} \mod p$

NOTICE: In the end, both parties have the same shared secret key $k = g^{ab} \bmod p$.

DH provides *Perfect Forward Secrecy*

If the secrets (a, b, k) of one communication session are compromised, then other (past or future) communication sessions will not be compromised.

The value of **forward secrecy** is that it protects past communications from being decrypted even if long-term keys are later compromised. As a result, the impact of a key compromise is limited: previously recorded traffic remains confidential, and only sessions established after the compromise are at risk until the key is revoked and replaced.

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Passive attacker:

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Why it cannot compute the shared secret key?

To build the shared secret key it needs to recover a (or b). One way of breaking DH, it to solve DLP, which is believed to be hard to break. However, there could other ways of breaking DH that do not necessarily need to solve DLP. However, after several years is still believed to be safe.

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Man-in-the-Middle attack:

1. Adversary fakes Bob identify and performs DH with Alice
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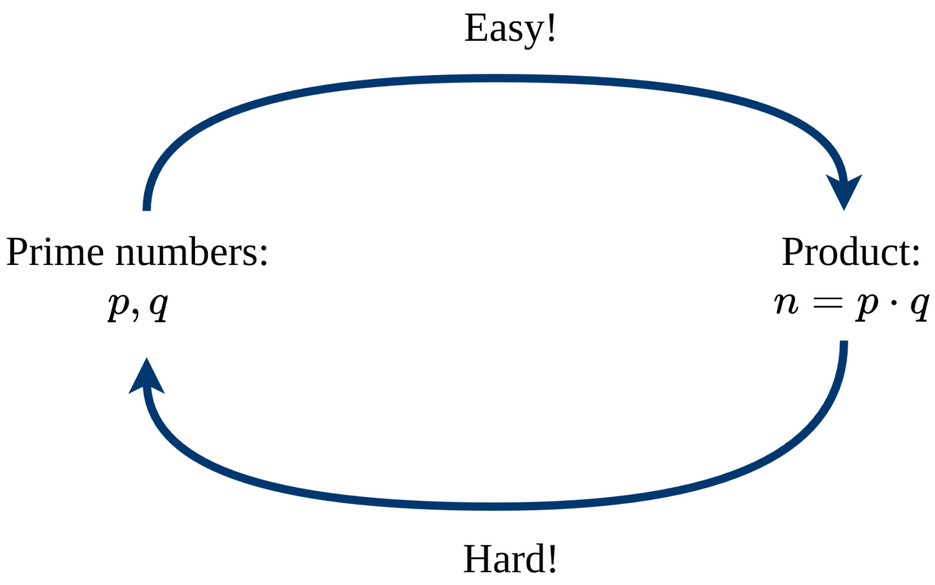
PROBLEM:
Alice and Bob do not ***authenticate*** the received public keys. We will see how to solve this problem when talking about authentication and the Public Key Infrastructure (PKI).

RSA (Rivest–Shamir–Adleman) cryptosystem

RSA

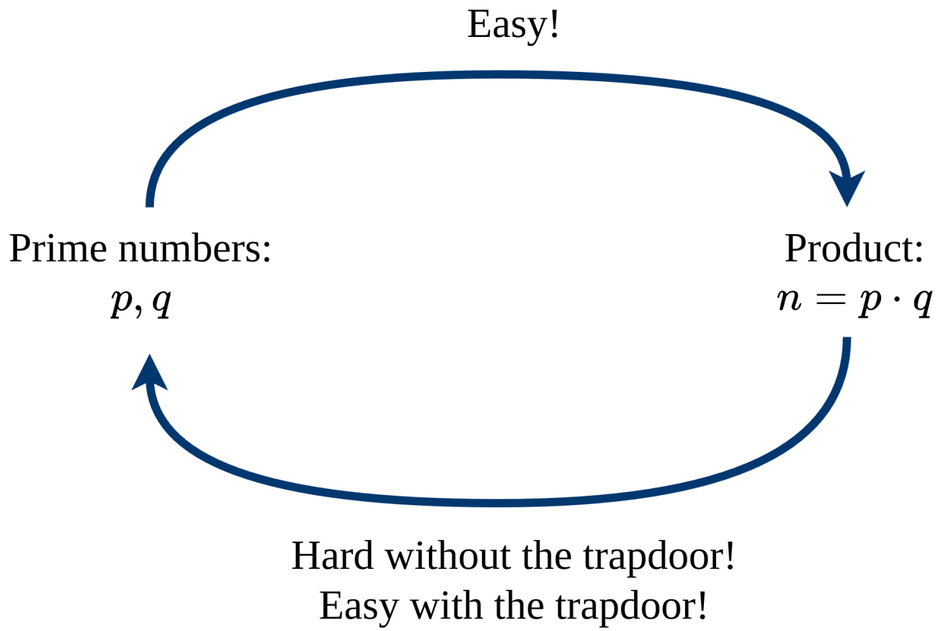
Public-key scheme proposed by Ron Rivest, Adi Shamir, and Leonard Adleman in 1977:



RSA is based on the ***integer factorization problem***.

Hard only when required

RSA exploits the concept of a ***trapdoor*** function:



The computation in the reverse direction is hard only when not knowing the trapdoor.

RSA Scheme

What we would like to have in principle:

- message: x
- public key: e
- private key: d (the trapdoor)
- scheme: $y = x^e$ such that $y^d = y^{ed} = x$
- work with modular arithmetic (to work easily with a finite set of numbers)

How to achieve this?

Euler's Phi Function (or Totient Function)

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When considering $\mathbb{Z}_m = \{0, 1, ..., m-1\}$

How many numbers in it are coprime (relatively prime) to m, i.e., $\Phi(m)$?

A naive approch to answer the previous question is to verify:

$$gcd(a, m) \stackrel{?}{=} 1 \; \forall \, a \in \mathbb{Z}_m$$

For instace, let us consider $\mathbb{Z}_6$:

$$\begin{aligned} gcd(0,6) = 6 &\Longrightarrow \text{✗} \\ gcd(1,6) = 1 &\Longrightarrow \checkmark \\ gcd(2,6) = 2 &\Longrightarrow \text{✗} \\ gcd(3,6) = 3 &\Longrightarrow \text{✗} \\ gcd(4,6) = 2 &\Longrightarrow \text{✗} \\ gcd(5,6) = 1 &\Longrightarrow \checkmark \\ gcd(6,6) = 6 &\Longrightarrow \text{✗} \end{aligned}$$

Hence:

$$\Phi(6) = 2$$

When considering other values for m:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
$\phi(n)$	1	1	2	2	4	2	6	4	6	4	10	4	12	6	8	8	16	6	18	8	12

Computing Efficiently the Euler's Phi Function

If we know how to factorize the number m:

$$m = p_1^{e_1} \cdot p_2^{e_2} \cdot \ldots \cdot p_k^{e_k}$$

where:

- $p_1, p_2, \ldots, p_k$ are distinct prime numbers
- $e_1, e_2, \ldots, e_k$ are positive integers

Then:

$$\Phi(m) = \prod_{i=1}^{m} (p_i^{e_i} - p_i^{e_i - 1})$$

$$m = 240$$

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$$m = 2^4 \cdot 3 \cdot 5 \Rightarrow \Phi(240) = (2^4 - 2^3)(3^1 - 3^0)(5^1 - 5^0) = 64$$

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$$\phi(p) = (p^1 - p^0) = p - 1$$

n is the product of two primes p and q

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$$\Phi(n) = \Phi(pq)\,?$$

$$\phi(n) = (p^1 - p^0)(q^1 - q^0) = (p-1)(q-1)$$

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4. Choose an integer e such that $1 < e < \phi(n)$ and e is coprime to $\phi(n)$:
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5. Compute d as the modular inverse of e modulo $\phi(n)$: $d \equiv e^{-1} \bmod \phi(n)$
   - we can use the Extended Euclidean algorithm to compute d

RSA: Private and Public Keys

1. (n, e) will be the public key, i.e., $k_{public} = (n, e)$

2. (n, d) will be the private key, i.e., $k_{private} = (n, d)$

RSA: Encryption

The encryption function is defined as:

$$y = E_{k_{public}}(x) = x^e \mod n$$

where x is the message, y is the ciphertext, (e, n) is the public key of the person receiving the message. Indeed, only the person with the private key (d, n) can decrypt the message.

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WARNING: As it must be, RSA use the public key of the receiver to encrypt the message!

RSA: Decryption

The decryption function is defined as:

$$x = D_{k_{private}}(y) = y^d = (x^e)^d = x^{ed} = x^1 = x \mod n$$

where x is the message, y is the ciphertext, (d, n) is the private key of the person <u>receiving</u> the message.

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NOTE: The following slides are painful. We know it. But they explain a bit the math behind RSA.

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To understand why it works, we need to introduce:

1. The Fermat's Little Theorem
2. The Chinese Remainder Theorem

Fermat's Little Theorem

Given a prime p, for any integer a, $a^p - a$ is a multiple of p, written as:

$$a^p \equiv a \bmod p$$

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Moreover if a and p are coprime then $a^{p-1} - 1$ is a multiple of p, written as:

$$a^{p-1} \equiv 1 \bmod p$$

PROOF: Check this video to see some examples that give you an intuition behind the proof of the theorem.

Chinese Remainder Theorem (CRT)

Given pairwise coprime positive integers $(n_1, n_2, \ldots, n_k)$ and an arbitrary integers $(a_1, a_2, \ldots, a_k)$, then the system:

$$\begin{aligned} x &\equiv a_1 \bmod n_1 \\ x &\equiv a_2 \bmod n_2 \\ &\vdots \\ x &\equiv a_k \bmod n_k \end{aligned}$$

has a unique solution modulo $n = n_1 \cdot n_2 \cdot \ldots \cdot n_k$.

PROOF: Check this video to see some examples that give you an intuition behind the proof of CRT.

NOTE: The practical implication of CRT is that if want to prove something when working with $x \mod n$ and $n = pq$, we can break the reasoning down into two parts: (1) $x \mod p$ and (2) $x \mod q$.

RSA: Proof of Correctness

From key generation:

$$ed \equiv 1 \bmod \Phi(n) \implies ed \equiv 1 \bmod (p-1)(q-1)$$

Thus:

$$ed \equiv 1 \bmod (p-1) \implies ed - 1 = k(p-1)$$

We prove correctness modulo (p) (similarly for (q)).

- If $\gcd(x, p) = 1$, then:

$$\begin{aligned} x^{ed} &\equiv x^{ed-1} \cdot x \equiv x^{k(p-1)} \cdot x \bmod p \\ &\equiv (x^{p-1})^k \cdot x \bmod p \\ &\equiv 1^k \cdot x \bmod p \quad \text{(Fermat's Little Theorem)} \\ &\equiv x \bmod p \end{aligned}$$

- Otherwise, it must be $\gcd(x, p) = p$ since p is prime, i.e., $x \equiv 0 \bmod p$, then:

$$x^{ed} \equiv 0 \equiv x \bmod p$$

By the CRT, $x^{ed} \equiv x \pmod{n}$.

RSA: Is it secure?

The main assumption behind RSA is that, given (e, n), it should be hard to compute d, p, q or $\Phi(n)$.

To make this hard:

1. p and q should two large (more than 1024 bits) distinct prime numbers
2. e should be equal or larger than 3
3. $(p-1)$ and $(q-1)$ should have large prime factors otherwise Pollard's rho algorithm is able to perform factorization efficiently

WARNING: Even when recommendations are followed, our *textbook* implementation of RSA suffers from different problems.

RSA: what if two moduli share a common factor p?

$$N_1 = p \cdot q_1$$
$$N_2 = p \cdot q_2$$
$$gcd(N_1, N_2) = p$$

Then we can easily recover q_1 and q_2:

$$q_1 = N_1/p$$
$$q_2 = N_2/p$$

RSA: avoid specific *corner case* messages

Messages such as x=0, x=1, x=(n-1) do not play well with exponentiation:

- $0^e = 0$
- $1^e = 1$
- if e is odd then: $(n-1)^2 = 1 \bmod n$, hence $(n-1)^e = (n-1) \bmod n$

RSA: small message, small exponent e

If x and e are small then:

$$y = x^e < n$$

Adversary can easily get the plaintext by computing:

$$x = \sqrt[e]{y}$$

RSA: same message encrypted with same exponent e

Given e ciphertexts of the same message x and the same exponent e is used during encryption but (p, q) different then the message x can be decrypted using the Chinese Remainder Theorem (CRT).

E.g., $e = 3$

$$\begin{aligned} y_1 &= x^3 \bmod n_1 \\ y_2 &= x^3 \bmod n_2 \\ y_3 &= x^3 \bmod n_3 \end{aligned}$$

CRT can build (here a step-by-step example):

$$y = x^3 \bmod (n_1 \cdot n_2 \cdot n_3)$$

And then $x = \sqrt[3]{y}$ since we know by construction that:

$$\begin{aligned} x &< \{n_1, n_2, n_3\} \\ x^3 &< n_1 \cdot n_2 \cdot n_3 \end{aligned}$$

RSA: same message encrypted with different coprime exponents

$$y_1 = x^{e_1} \bmod n$$
$$y_2 = x^{e_2} \bmod n$$

If we have that:

$$gcd(e_1, e_2) = 1$$

Then using EEA we can compute:

$$s \cdot e_1 + t \cdot e_2 = 1$$

But then:

$$y_1^s \cdot y_2^t = (x^{e_1})^s \cdot (x^{e_2})^t = x^{s \cdot e_1 + t \cdot e_2} = x$$

Here a step-by-step example.

RSA: small message space

Given a ciphertext y, if the message space is small (e.g., k values), then adversary can encrypt all k values and then compare results with the ciphertext y. Textbook RSA has no randomization.

RSA: How to mitigate/solve most of these attacks?

Textbook RSA is unsafe. In practice, we should always use a standard that provides rules on how to use a PK scheme for real-world usages:

> **SOLUTION**: PKCS (Public Key Cryptography Standards)

For instance:

- PKCS#1: RSA Cryptography Standard
- PKCS#3: Diffie Hellman Key Agreement Standard
- PKCS#11: Elliptic Curve Cryptography Standard

They define how to pad the messages, how to properly generate and validate the keys, etc.

How the break improper RSA uses?

Some tools that can be valuable:

1. **RsaCtfTool**: utility designed to decrypt data from weak public keys and attempt to recover the corresponding private key.

2. **msieve**, **yafu**, **cado-nfs**: tools for factoring large numbers.

This **blog post** provides examples (.pem files, i.e., a format for storing public/private keys) for testing known attacks.